An IoT-enabled Waste Management Solution

ESP32 • Automation • Wi-Fi

EcoSort – A smart dustbin

with Automatic Waste Segregation System

DRY · WET · METAL

Developer	Md Mizgan Ali
Platform	ESP32
Language	C++
Connectivity	Wi-Fi
Category	IoT & Embedded Systems
Application	Smart Waste Management

SYSTEM OVERVIEW

Specifications

CORE PARAMETERS

Controller	ESP-32 (DEV Kit V1)
Input Voltage	5–12 V DC
Actuator	Servo Motor
Segregation Type	Dry / Wet / Metal
Operation	Automatic lid opening & waste sorting
Power Supply	Power Module + Buck Converter

SENSORS USED

RD

Rain-Drop Sensor

Detects moisture — identifies wet waste

MD

Metal Detection Sensor

Inductive proximity type

IR

IR Sensor

Detects incoming objects at the inlet

WHERE IT FITS

Applications

SH

Smart Homes

Automatic waste segregation at the household level, with zero manual sorting effort.

MU

Municipalities

Helps local bodies manage garbage efficiently across public spaces and streets.

RC

Recycling Industries

Assists recyclers by pre-segregating waste before it reaches processing plants.

COMPLIANCE

Certifications Required

BIS

BIS Certification

Electrical safety compliance in India

CE

CE Certification

Safety & compliance for electronic goods

RoHS

RoHS Compliance

Restriction of hazardous substances — eco-friendly electronics

ISO

ISO 9001

Quality assurance for the manufacturing process

IEC

IEC Standards

Ensures safety & performance of electronic components

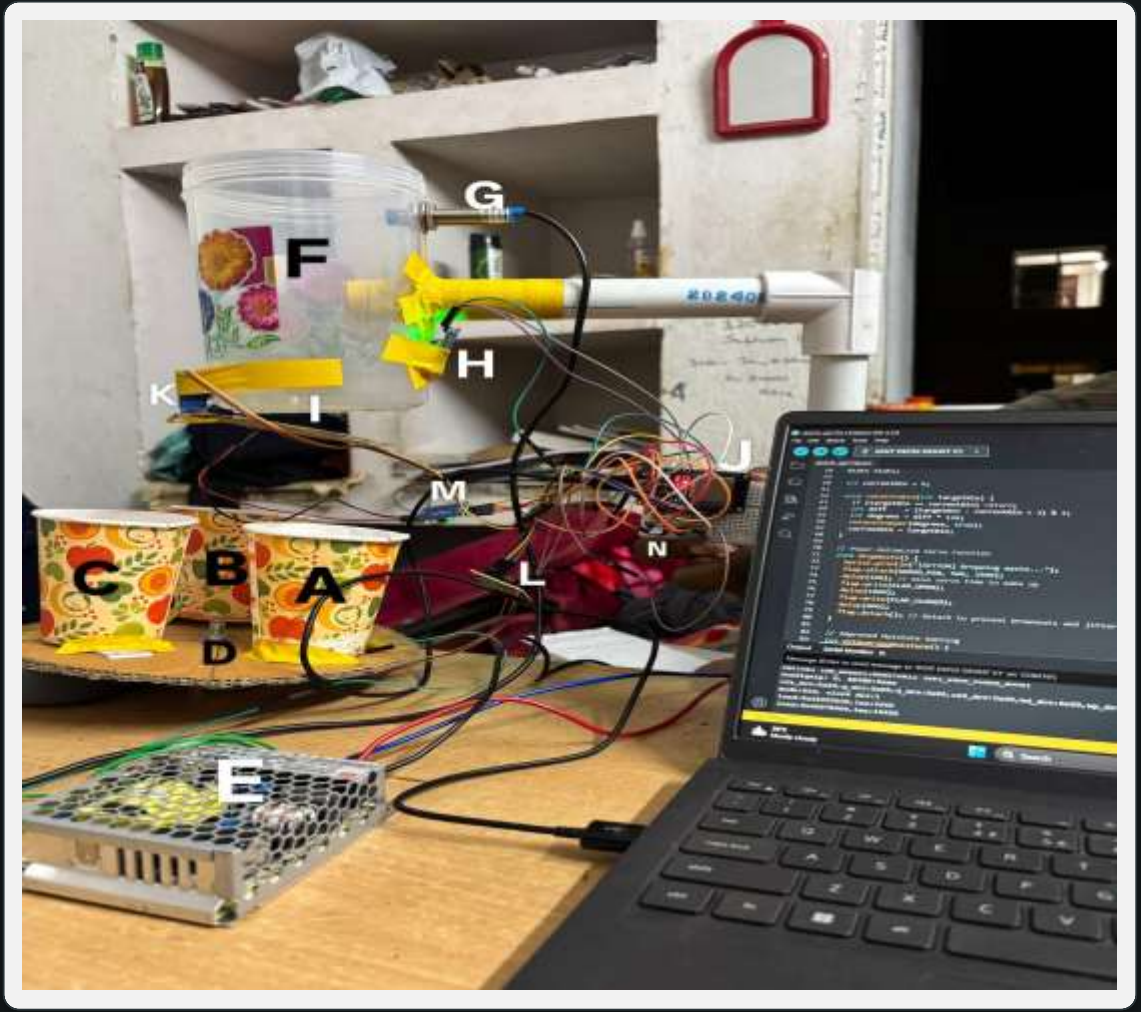
BILL OF MATERIALS

Components Required

Item	Key Specification	Qty	Manufacturer
ESP32 Dev Kit V1	Main controller board	1	Espressif Systems
Raindrop Sensor	3.3–5V · Analog & Digital out · LM393 IC	1	Generic (FlyRobo / SunRobotics)
Metal Detector Sensor	5–12V DC · Digital out · Inductive, ~2–10mm range	1	Omron
IR Sensor	3.3–5V · Digital out · ~2–30cm range	1	Excelitas Technologies
Servo Motor	4.8–6V · ~1.8 kg·cm torque · 0°–180°	1	TowerPro
Motor Driver	L298N	1	Generic
Power Module	100–240V AC in · 36V, 1.45A out · 50W	1	Generic
Jumper Wires	Connecting wires	As needed	Generic
Breadboard	~400–830 point · 5–12V use	1	Generic
Dustbin Body	Plastic enclosure	1	Local

Prototype Overview

- **A** Metal Waste Bin
- **B** Dry Waste Bin
- **C** Wet Waste Bin
- **D** 5mm Shaft Adapter
- **E** Power Module
- **F** Main Dustbin
- **G** Proximity Metal Sensor
- **H** IR Sensor
- **I** Raindrop Sensor
- **J** ESP32 Devkit V1
- **K** Servo Motor
- **L** Stepper Motor Driver
- **M** Raindrop Sensor Driver
- **N** 100µF Capacitor



HOW IT WORKS

Core Components

ESP32 — Main controller — processes every sensor signal and drives the servo; built-in Wi-Fi/Bluetooth enables future IoT monitoring.

RD — Raindrop Sensor — measures moisture via conductivity changes to classify waste as wet.

MD — Metal Detector — uses electromagnetic induction to route metallic objects to the metal bin.

IR — IR Sensor — detects an inserted object and triggers the segregation sequence.

SV — Servo Motor — rotates the internal flap to guide waste into the correct compartment.

ST — Stepper Motor — divides rotation into precise steps for accurate positioning, paired with a driver.

LESSONS LEARNED

Development Challenges

01
Sensor Calibration

Setting proper sensitivity for moisture & metal sensors was difficult, leading to occasional misclassification.

02
False Detection

Dry and wet waste sometimes produced similar readings, causing segregation errors.

03
Servo Alignment

Positioning the servo and flap mechanism accurately for correct waste direction was challenging.

04
Circuit Complexity

Managing multiple sensors and connections with the ESP32 required careful wiring and debugging.

05
Power Supply

Ensuring stable power for all components was difficult, especially during simultaneous operation.

06
Physical Design

Designing a compact, efficient multi-compartment dustbin structure was not easy.

WHAT'S NEXT

Scope of Improvement

IoT

IoT Integration

Monitor bin status remotely via ESP32 Wi-Fi and a mobile app.

SNS

Better Sensors

Ultrasonic or image-based detection for improved classification accuracy.

AI

AI-Based Segregation

Machine learning or camera modules to identify waste more precisely.

PWR

Power Optimization

Battery backup or solar power for greater energy efficiency.

ODR

Odor Control

Deodorizing / air-filter systems to improve hygiene.

CMP

Auto Compaction

A compression mechanism to increase effective storage capacity.

COMPETITIVE LANDSCAPE

Similar Products in the Market

01	**Automatic Sensor Dustbins**	Use IR sensors to open lids automatically, but perform no waste segregation.
02	**Smart Segregation Bins**	Advanced bins that separate dry and wet waste using basic sensor technologies.
03	**AI-Based Smart Bins**	Use cameras and artificial intelligence to identify and sort waste accurately.
04	**Industrial Segregation Systems**	Large-scale machinery used in recycling plants for automatic sorting.
05	**IoT-Enabled Smart Bins**	Connected bins that monitor fill levels and send timely collection alerts.

GO TO MARKET

Marketing Strategy

Target Audience

- Smart homes
- Schools & colleges
- Municipal corporations

Promotion Methods

- YouTube demonstrations
- Social media marketing
- Eco-campaign collaborations
- Exhibitions & science fairs

Selling Platforms

- Amazon / Flipkart
- Local electronics stores
- Direct institutional sales

Key Selling Points

- Promotes a clean environment
- Saves manual effort
- Supports recycling

BUDGET

Cost Breakdown

Component	Unit Price (₹)	Qty	Cost (₹)
Proximity Sensor (PNP NO)	309	1	309
IR Sensor	143	1	143
Stepper Motor	349	1	349
Raindrop Sensor	119	1	119
5mm Shaft Adapter	79	1	79
Servo Motor	139	1	139
Buzzer	79	1	79
ESP32 Development Board	521	1	1042

TOTAL PROJECT COST

₹1,911

A low-cost, modular build using off-the-shelf sensors and an ESP32 controller — keeping the prototype affordable for institutional and home use.

Thank You!

Building a Cleaner Tomorrow with Smart Automation.

Credentials:-
Md Mizgan Ali
mizganali12@gmail.com
https://github.com/mizgan786